“On the way with Python”

Lecture

04

Python Data Types

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Introduction

In Python, data types define the type of value a variable can store. Python is a dynamically typed language, which means you do not need to declare the data type explicitly. The interpreter automatically assigns the data type based on the value.

Why Data Types are Important

Data types are important because they:

- Define what kind of data a variable holds
- Decide which operations can be performed
- Help prevent errors in programs
- Improve program readability and logic

Data Types

Data Type	Example
int	10
float	10.5
str	"Hello"
bool	True / False
list	[1, 2, 3]
tuple	(1, 2, 3)
dict	{"name": "John"}
set	{"red", "green", "blue"}

Mutable vs Immutable Data Types

- **Mutable Data Types:** These can be changed after creation
- **Immutable Data Types:** These cannot be changed after creation.

Mutable	Immutable
list	int
dict	float
set	bool
	tuple
	str

Checking Data Types

- Use the type() function to check a variable's data type.

```
x = 10
print(type(x))  # <class 'int'>
```

Type Conversion (Type Casting)

- Converting one data type to another.

```
x = "10"
y = int(x)

price = float(5)
text = str(100)
```

""Thanks All""